EECS 498/598 Final Project Report

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ML for Symptom Classification Based on Cough Audio Signals

Project Statement

Respiratory conditions, such as the cold and flu, are widespread, leading to a significant need for accessible monitoring tools. Smartphones are sensing platforms that are almost always available to users. The goal of our project is to develop a non-diagnostic screening tool capable of real-time cough pattern classification, identifying categories like healthy, symptomatic, or COVID-19. This system prioritizes user privacy by ensuring that all raw audio processing is performed locally on the device, rather than relying on the cloud, as many existing systems do.

Project Challenges

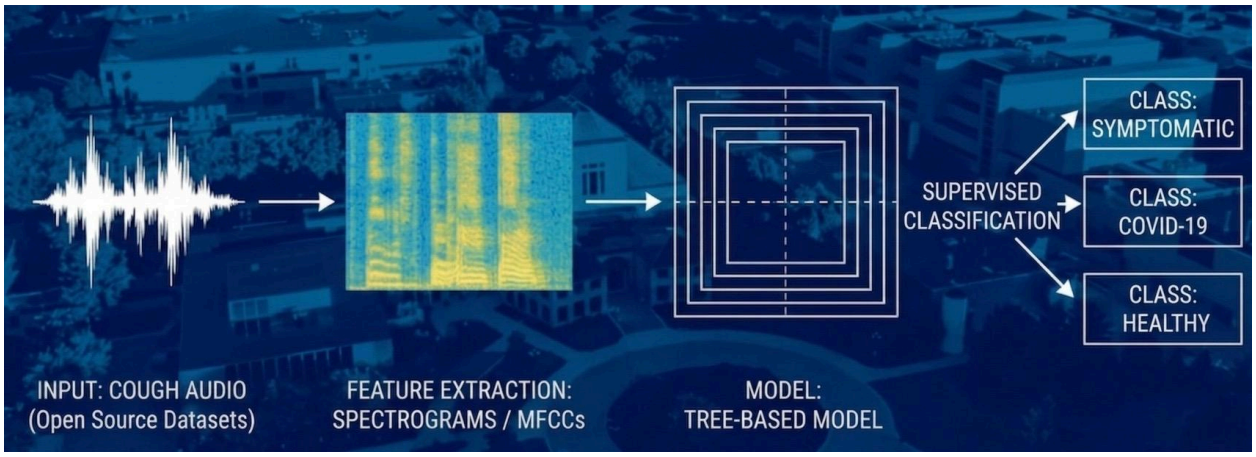
One of the biggest challenges in our project was the quality and variability of the COUGHVID dataset. Recordings differ widely in duration, classification quality, and background noise because they were collected from many different sources. There are many recordings with no cough recorded at all. This variability likely contributes to lower classification accuracy.

Embedded Systems Component

The embedded systems component focuses on building a real-time, on-device inference pipeline that uses the device microphone to capture cough audio and classify it locally. The system records a short audio segment, preprocesses it, converts it into a feature representation, and passes those features into the trained machine learning model. Our pipeline integrates sensing, signal processing, and inference directly on a mobile or embedded platform without relying on cloud computation.

A key part of this component is keeping the pipeline efficient enough to run on resource-constrained hardware such as a smartphone. To support this, the device first captures cough audio and converts it into a compact representation using Mel spectrograms and related audio features. These features are then used by the trained model for classification. Because the system runs entirely on-device, it offers privacy advantages and reduced dependence on network connectivity.

Machine Learning Component



For the machine learning component, we use the publicly available COUGHVID dataset, which contains cough recordings collected from a wide range of users and environments. Our task is to classify recordings such as healthy, symptomatic, and COVID-19-related coughs. Since cough audio is naturally represented as time-frequency information, our pipeline begins by extracting spectrogram-based features from the recordings.

Initially, we experimented with traditional approaches using flattened spectrogram representations as input to a classifier. However, to improve performance while still maintaining efficiency for device deployment, we changed the primary model to an XGBoost forest. XGBoost is a gradient-boosted decision tree method that is often effective on structured feature representations and can deliver a better balance between accuracy, training efficiency, and deployment practicality than more complex deep learning models.

In the pipeline, each cough recording is converted into features derived from spectrograms and Mel spectrograms. Raw audio signals are preprocessed and transformed into feature vectors containing MFCCs, chroma features, and spectral contrast to capture essential temporal and frequency data. These extracted features are then provided to the XGBoost model for classification. Compared with earlier approaches, this method avoids the higher computational cost of deep neural networks while still capturing useful acoustic information. It is also more compatible with embedded deployment because tree-based models can be easier to store and execute efficiently on limited hardware.

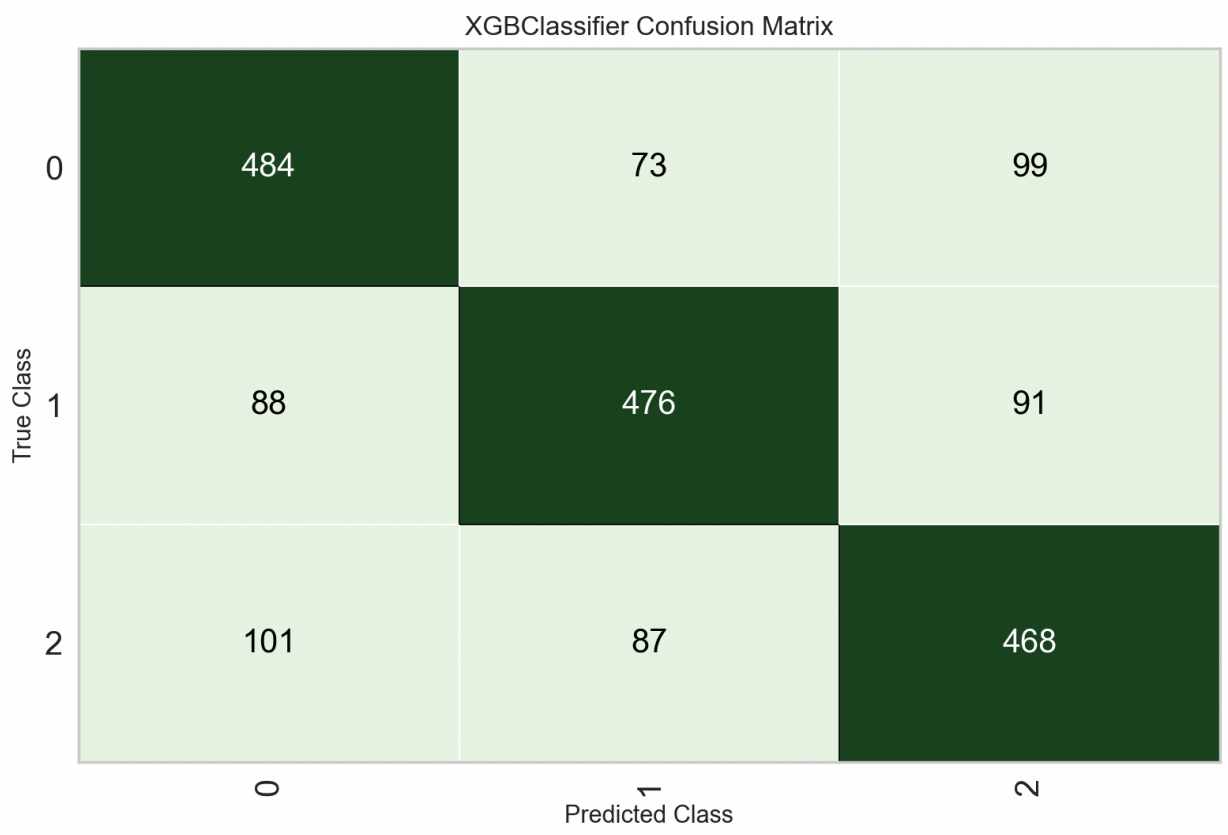
We also used PyCaret, a machine learning framework and Python library that makes deploying and training our model and preprocessing the data simpler. It was particularly useful for us when we were figuring out what kind of model would be best-suited for our project. PyCaret allowed us to quickly test our data with different classification models, evaluate performance, and identify the strongest models. We also used PyCaret to visualize performance similar to our previous projects and assigned readings in this course (confusion matrices and ROC curves).

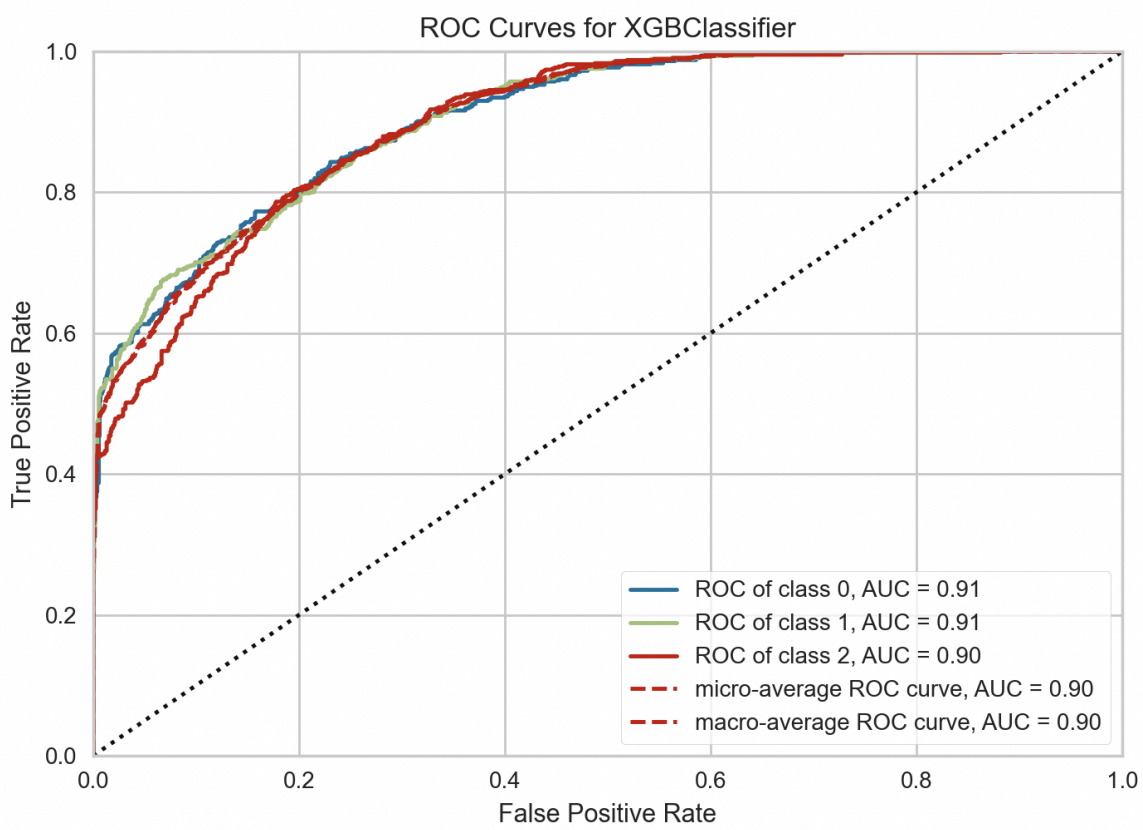
Putting it Together

The final end-to-end pipeline moves from microphone input to cough segmentation to feature extraction and, finally, to an XGBoost classification. This integration of sensing and machine learning allows for immediate decision-making directly on the device. The processing flow is designed to be lightweight, allowing for quick input generation, while still maintaining privacy-preserving principles. This design ensures the system stays functional within the constraints of the mobile and embedded environments.

Implementation and Evaluations

Our evaluations indicate high performance, with AUC scores reaching 0.90 to 0.91 across various symptom classes. We have attached our visualizations below (0 = COVID, 1 = HEALTHY, 2 = SYMPTOMATIC). Real-time implementations have also demonstrated the system's ability to successfully identify healthy versus symptomatic patterns. However, there are still various limitations to investigate. For example, how to safely collect “unhealthy” real-world data, like COVID-19 cough samples, and minimize variability across many device microphones should still be explored. Future work on our pipeline may include model compression, noise-robust features, and the exploration of lightweight CNNs for more nuanced pattern recognition.


XGBClassifier Confusion Matrix
True Class
Predicted Class
0
1
2
484
73
99
88
476
91
101
87
468


ROC Curves for XGBClassifier
True Positive Rate
False Positive Rate
0.0
0.2
0.4
0.6
0.8
1.0
ROC of class 0, AUC = 0.91
ROC of class 1, AUC = 0.91
ROC of class 2, AUC = 0.90
micro-average ROC curve, AUC = 0.90
macro-average ROC curve, AUC = 0.90

Individual Contributions

Mary Czelusniak : 100%

Pooja Vinukonda : 100%

Divya Pothavajhyula : 100%

We all worked on all parts of the project together, so all of our contributions are 100%.

Source Code (with Documentation – refer to README.md) and Video Demo

We have linked our GitHub repository below, which contains the source code with documentation and our video demos.

https://github.com/divyabpoth/598-proj

Limitations and Future Work

One constraint is regarding environmental noise, as distinguishing cough signals from background elements, such as television or conversation, still remains a challenge. Current evaluations are also limited by a lack of real-world "unhealthy" data, so we had to rely on curated datasets and playback rather than live symptomatic samples. To address these, future work will focus on incorporating noise-robust features and expanding the training data with more diverse, real-world recordings. The team plans to improve the latency-accuracy trade-off through model compression and pruning. We would also like to explore lightweight CNNs or hybrid architectures to capture finer patterns that traditional feature-based models might overlook. Lastly, we want to ensure that our system can generalize across devices. This will require testing on varying microphones and microphone qualities (various smartphone and laptop microphones) and the application of advanced normalization techniques. The system's performance may also vary based on the distance between the user and the microphone. This is because changes in amplitude and signal quality can impact our feature extraction and classification accuracy. We will evaluate how distance from the microphone affects performance by recreating our testing at various distances to account for many different recording conditions.

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Agenda

- Project Statement
- Project Challenges
- Embedded Systems Component
- Machine Learning Component
- Putting Them Together
- Implementation and Evaluations
- Video Demo
- Limitations and Future Work

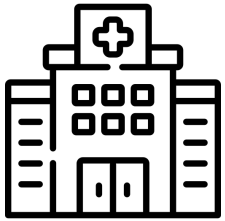
Project Statement

- Respiratory conditions (e.g., cold, flu) are common & widespread
- Smartphones are ‘always’ available sensing platforms
- Existing solutions rely on cloud processing

Need: On-device, non-invasive, privacy-preserving cough analysis

Goal: Real-time cough pattern classification (non-diagnostic)

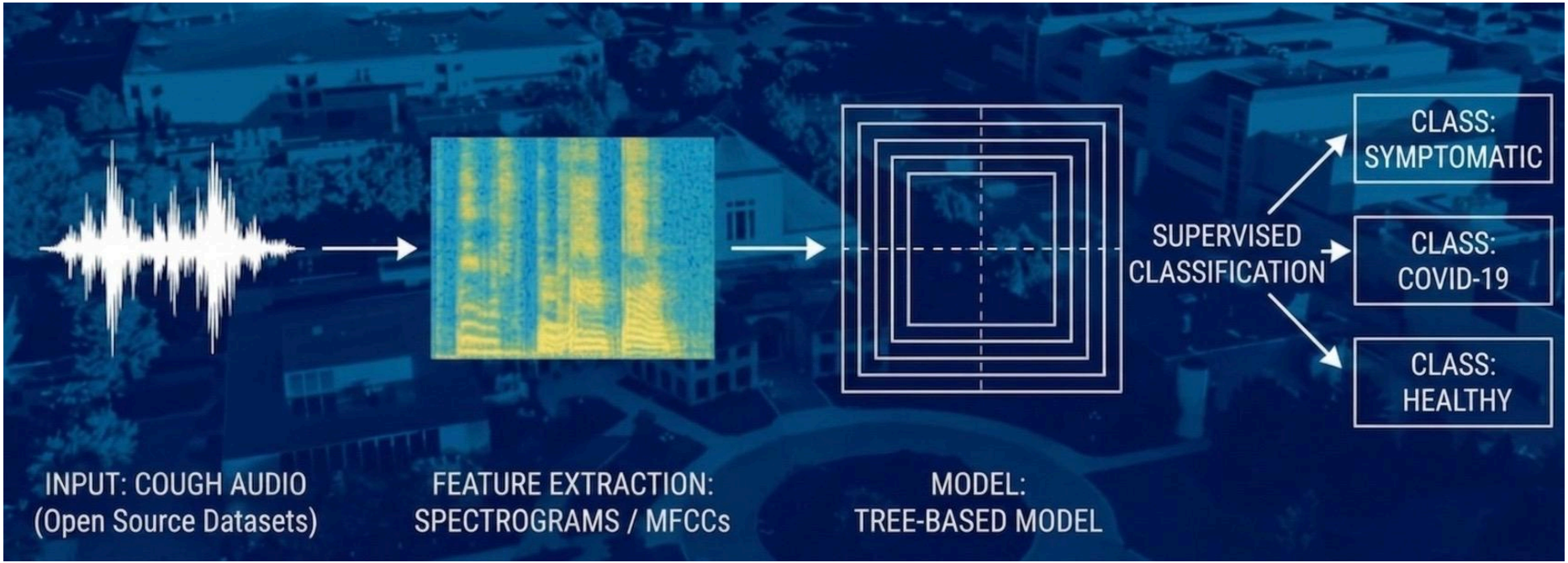
- Target classes: healthy vs. symptomatic vs. COVID-19



Project Challenges

- Data Quality & Preprocessing
- Cough Segmentation
- Feature Representation
- Safe Testing of “Unhealthy” Cases
- On-Device Constraints

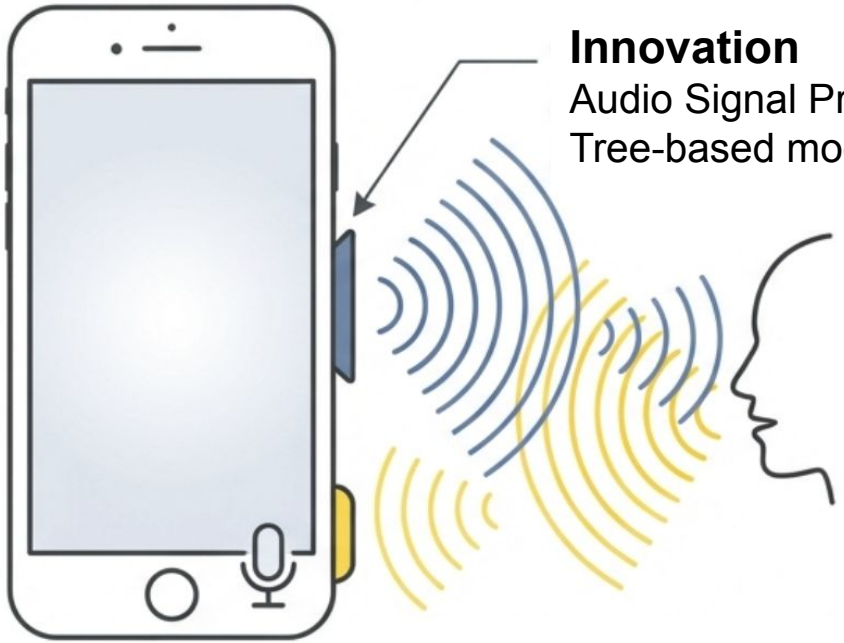
Healthy vs. Symptomatic


CLASS:
SYMPTOMATIC
SUPERVISED
CLASSIFICATION
CLASS:
COVID-19
CLASS:
HEALTHY
INPUT: COUGH AUDIO
(Open Source Datasets)
FEATURE EXTRACTION:
SPECTROGRAMS / MFCCs
MODEL:
TREE-BASED MODEL

Computational Sensors & Device Setup

Platform
Smartphone (or laptop)
System constraints: energy, compute, memory, latency

Innovation
Audio Signal Processing
Tree-based models



Primary Sensor
Built-in microphone
Real-time audio capture

Fully On-Device Pipeline: Real-time audio capture → feature extraction → inference

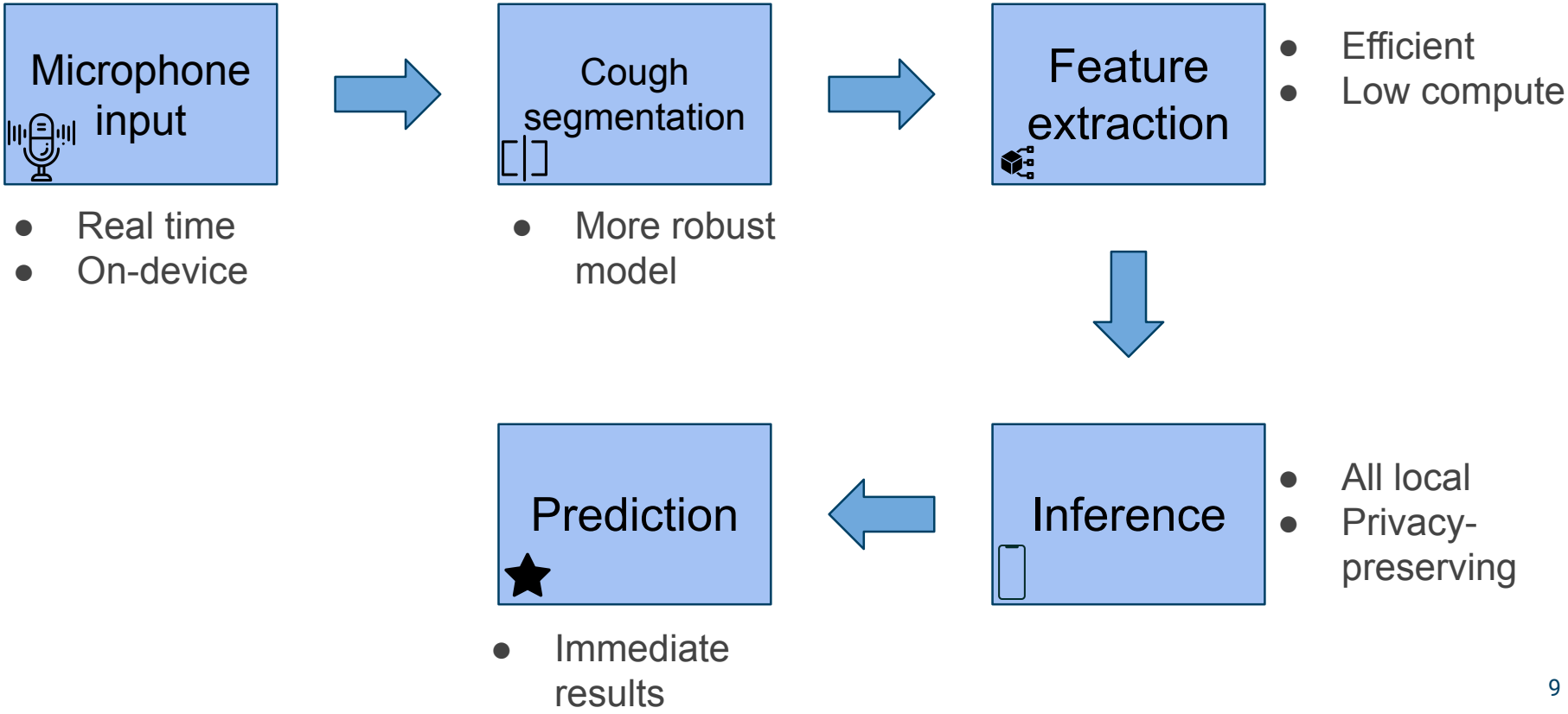
Embedded Systems Component

- **On-Device Audio Sensing:** Utilizing the device microphone to capture real-time cough audio signals
- **Local Real-Time Inference Pipeline:** Recording short audio segments and processing them immediately for classification and running a trained XGBoost model directly on-device without cloud support
- **Signal Processing on Device:** Converting audio into Mel spectrogram representations suitable for model input

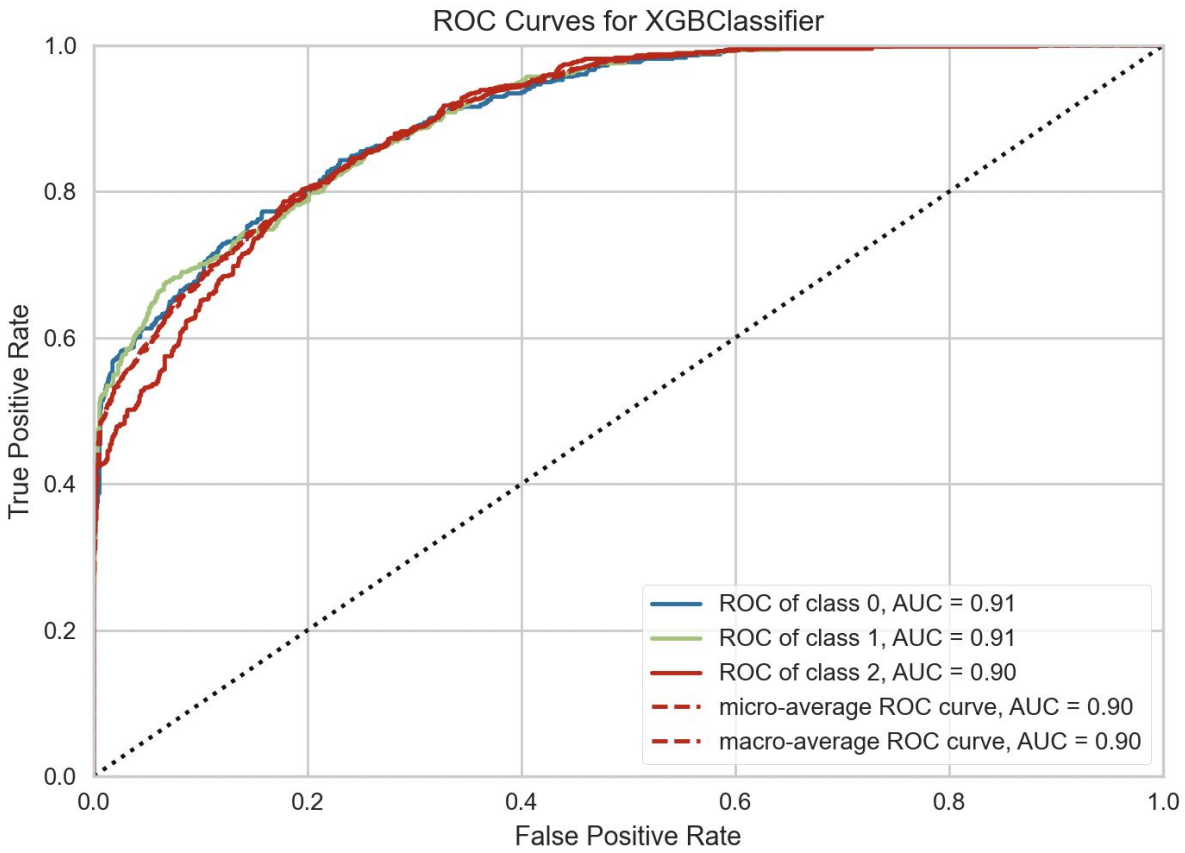
ML Component

- **Classification Task:** categorize coughs into *healthy*, *symptomatic,* or *covid-19*
- **Dataset:** COUGHVID from Kaggle
  - 25K+ recordings → filtered to include only ‘good quality recordings’
  - Used a balanced subset
  - Isolated individual coughs
  - Extracted features to capture temporal / frequency characteristics
- **Model:** XGBoost via PyCaret library
- **Optimization:** Normalization + feature transformations → balance classification accuracy with computational efficiency

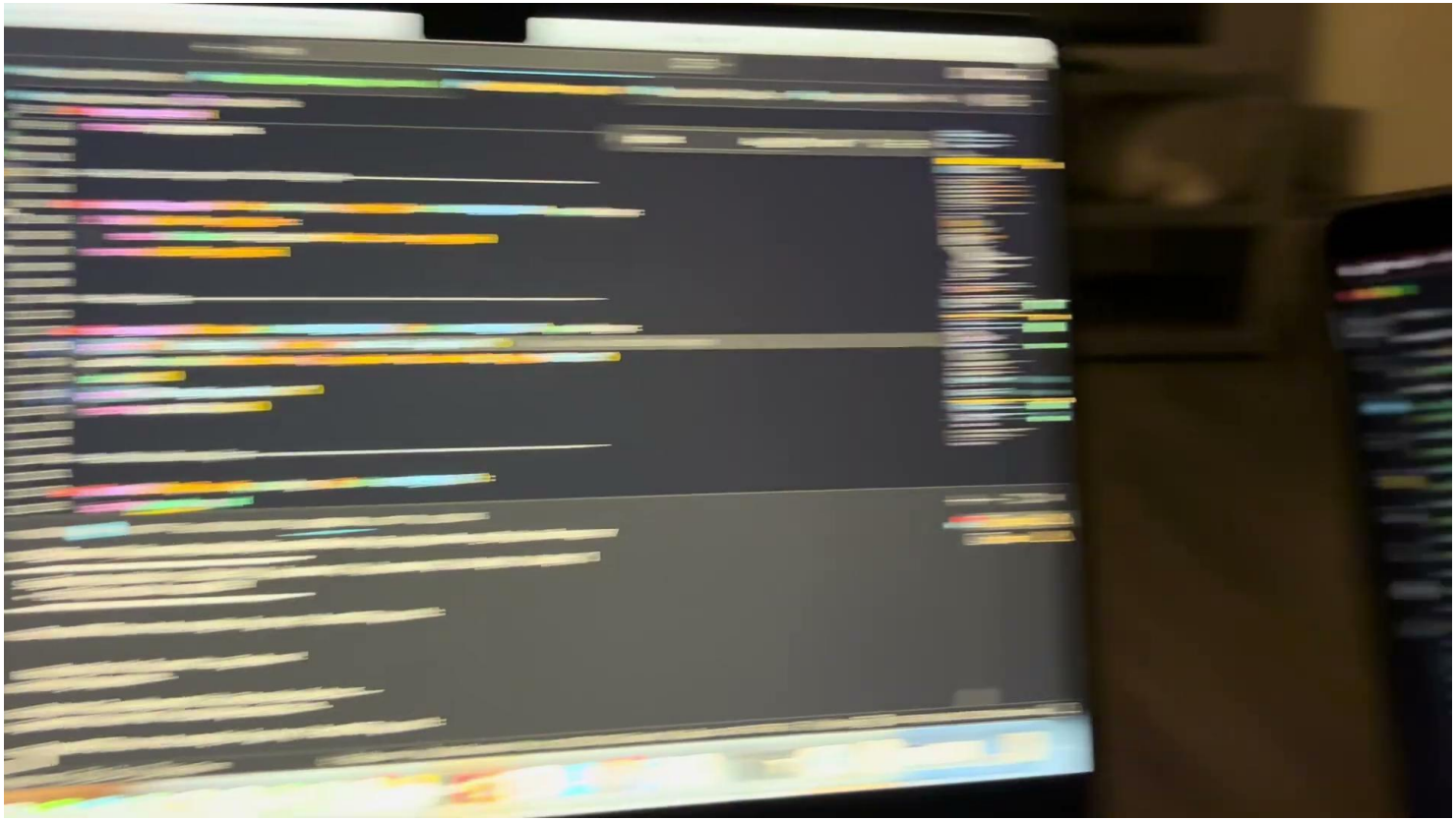
Putting Them Together: End-to-End Pipeline



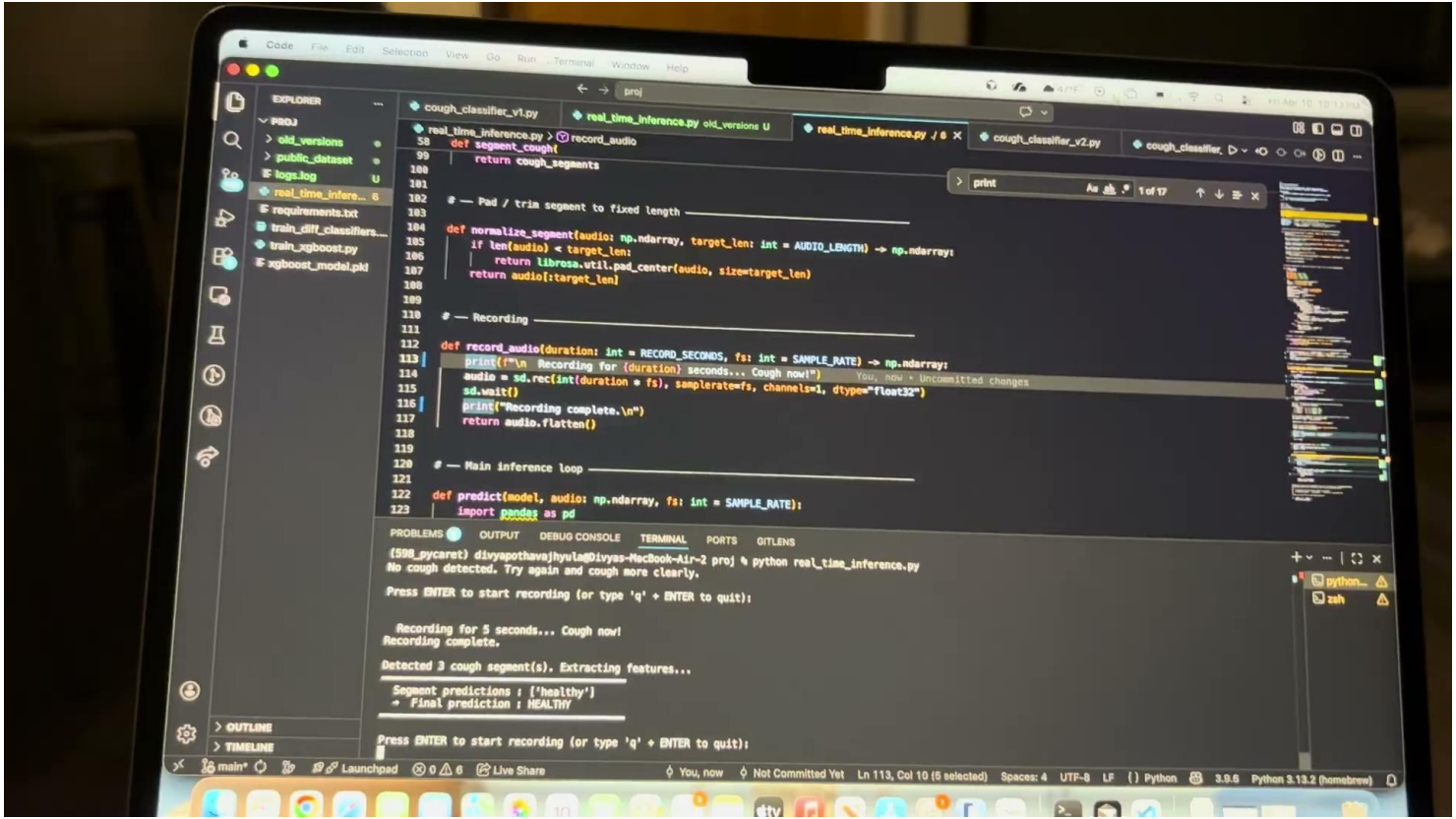
Implementations/Evaluations



Video Demo: COVID-19



Video Demo: Healthy



Limitations & Future Work

- **Environmental Noise:** Difficulty distinguishing cough signals from background noise (e.g., conversations, TV) in real-world settings
  → **Future Work:** Incorporate noise-robust features and filtering techniques
- **Real-Time Latency vs. Accuracy:** Trade-off between feature extraction complexity and fast on-device inference
  → **Future Work:** Optimize feature pipeline and explore model compression/pruning
- **Limited Real-World "Unhealthy" Data:** Challenges in safely collecting live unhealthy cough samples; reliance on curated datasets or playback
  → **Future Work:** Expand dataset with more diverse, real-world recordings
- **Generalization Across Devices:** Variability in microphone quality and recording conditions may impact performance
  → **Future Work:** Test across devices and apply normalization/adaptation techniques

Discussion & Feedback